

OC IT Projects — Chapter 5

Quiz + Quick Review

Focus: Ethernet, switching, domains, switch features, VLANs, and network security.

1. Quick Review

- CSMA/CD → Carrier Sense Multiple Access / Collision Detection; the device checks the medium before transmitting and collisions can occur when transmissions overlap.
- Collision Domain → group of devices that can potentially experience collisions. A switch separates collision domains by ports.
- Broadcast Domain → group of devices receiving a broadcast; the training's switch example shows multiple collision domains inside one broadcast domain.
- Hub → broadcasts received data to connected devices.
- Bridge → presented as an older version of a Switch.
- Switch → forwards traffic toward the intended destination and improves performance by separating collision domains.
- Router → connects two or more networks and determines the next network point.
- Ethernet speed / Cable Categories → cable categories are relevant to transmission capabilities.
- 802.3ad → Link Aggregation.
- 802.3af / 802.3at → Power over Ethernet (PoE).
- Port Mirroring → copies traffic for monitoring/analysis.
- 802.1X → port-based network access authentication.
- MAC Filtering → access control based on MAC addresses.
- QoS → managing/prioritizing network traffic.
- VLAN → logical separation into different broadcast domains.
- 802.1Q → VLAN Trunking.
- IDS/IPS → detecting and/or preventing suspicious or malicious network activity.
- NGFW → Next-Generation Firewall used to inspect/control network traffic.

2. Chapter 5 — Full Quiz

30 questions. Answer first, then check the key.

Q1. What does CSMA/CD stand for?

- A) Carrier Sense Multiple Access / Collision Detection
- B) Carrier System Multiple Access / Cable Detection
- C) Computer Sense Multiple Access / Collision Domain
- D) Carrier Signal Management / Collision Detection

Q2. Before transmitting data using CSMA/CD, what does a device do?

- A) Checks whether the medium is available
- B) Changes its IP address
- C) Contacts the router
- D) Creates a VLAN

Q3. Two devices transmit at the same time on a shared medium. What can occur?

- A) VLAN
- B) Collision
- C) Trunk
- D) Port mirroring

Q4. What is a Collision Domain?

- A) A group of devices that can potentially experience collisions with each other
- B) A group of devices connected to different routers
- C) A group of VLANs
- D) A group of servers

Q5. What is the effect of a Switch on Collision Domains?

- A) It combines all devices into one collision domain
- B) It creates separate collision domains per switch port
- C) It eliminates broadcast traffic
- D) It converts collision domains into VLANs

Q6. How does a Hub handle received data?

- A) Sends it only to the destination
- B) Broadcasts it to the connected devices
- C) Routes it to another network
- D) Encrypts it

Q7. Why does a Switch generally provide better performance than a Hub?

- A) It sends traffic to the intended destination instead of broadcasting everything to all ports
- B) It always uses fiber
- C) It eliminates IP addresses
- D) It works only with routers

Q8. According to the training, a Bridge is essentially:

- A) A newer version of a router
- B) An older version of a Switch
- C) A type of firewall
- D) A type of transceiver

Q9. What is the primary purpose of a Router?

- A) Connect two or more networks

- B) Connect only computers inside one LAN
- C) Broadcast frames to all ports
- D) Provide PoE

Q10. Which device is primarily used to connect devices within a LAN and forward traffic to the appropriate destination?

- A) Router
- B) Switch
- C) Firewall
- D) Hub

Q11. Scenario: PC1 sends data to PC2 through a Hub. What happens?

- A) Only PC2 receives the transmission
- B) The transmission is sent to the connected devices
- C) The router receives it first
- D) A VLAN is automatically created

Q12. Scenario: You replace the Hub with a Switch. What is the major improvement?

- A) Traffic can be forwarded toward the intended destination rather than being broadcast to every connected device
- B) All devices become one broadcast domain per port
- C) IP routing becomes unnecessary
- D) The Switch becomes a firewall

Q13. What can a Switch create?

- A) Separate Collision Domains
- B) Separate Internet connections
- C) Separate DNS servers
- D) Separate IP protocols

Q14. A Switch has four active ports connected to four devices. According to the training's example, how many Collision Domains can there be?

- A) 1
- B) 2
- C) 4
- D) 8

Q15. In the Switch example from the training, how many Broadcast Domains are shown?

- A) 1
- B) 2
- C) 4
- D) 8

Q16. Which topic is associated with Ethernet speed limitations in the training?

- A) Cable Categories
- B) DNS Records
- C) DHCP DORA
- D) Firewall Rules

Q17. Why are cable categories important when discussing Ethernet speeds?

- A) Different cable categories support different transmission capabilities
- B) They determine the IP address
- C) They determine the VLAN ID

D) They replace switches

Q18. What is Link Aggregation associated with?

- A) 802.3ad
- B) 802.1X
- C) 802.1Q
- D) 802.3af

Q19. What is the purpose of Link Aggregation?

- A) Combine multiple physical links to improve capacity/redundancy
- B) Encrypt all network traffic
- C) Assign IP addresses
- D) Create DNS records

Q20. PoE stands for:

- A) Power over Ethernet
- B) Protocol over Ethernet
- C) Port over Encryption
- D) Power of Equipment

Q21. Which standards are associated with PoE in the training?

- A) 802.3af / 802.3at
- B) 802.1Q / 802.1X
- C) 802.3ad / 802.11ac
- D) 802.1D / 802.1w

Q22. Scenario: An Ethernet-connected IP device also needs electrical power through the Ethernet connection. Which feature is needed?

- A) PoE
- B) QoS
- C) Port Mirroring
- D) MAC Filtering

Q23. What is Port Monitoring / Mirroring used for?

- A) Copying traffic from a port for monitoring/analysis
- B) Providing electrical power
- C) Assigning VLAN IDs
- D) Connecting two networks

Q24. Which statement best describes 802.1X in the training?

- A) Port-based network access authentication
- B) Ethernet power delivery
- C) Link aggregation
- D) VLAN trunking

Q25. Which feature can be used to control access based on MAC addresses?

- A) MAC Filtering
- B) Port Mirroring
- C) Link Aggregation
- D) PoE

Q26. What is QoS primarily concerned with?

- A) Managing/prioritizing network traffic
- B) Supplying electrical power

- C) Assigning MAC addresses
- D) Creating physical cables

Q27. What is the main idea behind a VLAN?

- A) Logically separate a network into different broadcast domains
- B) Increase cable length
- C) Replace the router completely
- D) Convert copper into fiber

Q28. Which standard is associated with VLAN Trunking?

- A) 802.3ad
- B) 802.3af
- C) 802.1Q
- D) 802.1X

Q29. Which statement best describes IDS/IPS?

- A) Used for detecting/preventing suspicious or malicious network activity
- B) Provides electrical power over Ethernet
- C) Aggregates Ethernet links
- D) Creates cable pinouts

Q30. Scenario: Your company needs a security device that can inspect and control network traffic, and the training specifically introduces Next-Generation Firewall. Which technology is this?

- A) NGFW
- B) PoE
- C) VLAN Trunking
- D) Link Aggregation

3. Answer Key

1-A | 2-A | 3-B | 4-A | 5-B

6-B | 7-A | 8-B | 9-A | 10-B

11-B | 12-A | 13-A | 14-C | 15-A

16-A | 17-A | 18-A | 19-A | 20-A

21-A | 22-A | 23-A | 24-A | 25-A

26-A | 27-A | 28-C | 29-A | 30-A

4. Quick Memorization

- CSMA/CD → Carrier Sense Multiple Access / Collision Detection.
- Hub → broadcasts received data to connected devices.
- Switch → forwards traffic toward the intended destination and separates Collision Domains.
- Bridge → older version of a Switch (training terminology).
- Router → connects two or more networks.
- 802.3ad → Link Aggregation.
- 802.3af / 802.3at → PoE.
- Port Mirroring → copies traffic for monitoring/analysis.
- 802.1X → port-based network access authentication.
- MAC Filtering → access control based on MAC addresses.
- QoS → traffic management/prioritization.
- VLAN → logical separation into different broadcast domains.
- 802.1Q → VLAN Trunking.
- IDS/IPS → detect/prevent suspicious or malicious network activity.
- NGFW → Next-Generation Firewall for inspection/control of network traffic.

Your Chapter 5 quiz result: 30/30 (100%).